

PCA

eigen decomp.

- cor.mat = cor(dat)
- eig = eigen(cor.mat)
- plot scree
- loading = eig\$vector
- scores = dat %>% %>% loading

reconstruction $\hat{X}$

A = cbind(loadings...)

Z = cbind(scores...)

X-reconst = Z %>% %>% %>% A

→ col-means = colMeans(dat)

→ col-sd = apply(dat, 2, sd)

→ reconstruct-unscaled = t(t(X-reconst) * col-sd + col-means)

SVD

svd-result = svd(dat)

loading = svd-result\$u

scores = dat %>% %>% loading

scores-alt = svd\$u %>% %>% diag(svd\$d)

Correlation between PC score and individual vars.

corPC1 = cor(score1, dat)

$$\rho_{Y_i, X_k} = \frac{\sqrt{\lambda_i} e_{ik}}{\sqrt{\sigma_{kk}}}$$

or: d1 = svd-result\$d[1]

u1 = svd-result\$u[,1]

cor-svd = u1 * (d1 / sqrt(nrow(dat)-1))

Fishers LDA

- 1.) Calculate group mean and overall mean.
- 2.) compute B
- 3.) compute W
- 4.) eig-mat = eigen(W⁻¹B) %>% %>% Re()
- 5.) calc Spoke so that a_k¹ Spoke a_k = 1 → W / (sum(n_i) - g)
- 6.) standardize eig-mat → eig-mat = (eig-mat %>% %>% diag(1/sd))[1,2]
- 7.) calculate scores: as.numeric(X %>% %>% eig-mat[,1])
as.numeric(X %>% %>% eig-mat[,2])
- 8.) Add scores to original data set then plot.

for Ci in 1:g

diff = group-means[Ci,] - overall-mean

B = B + n_i * (diff %>% %>% t(diff))

X_i = X[dat\$group == i,]

W = W + cov(X) * (nrow(X)-1)

}

Regression

Formula

$$Y = \text{cbind}(\dots)$$

$$X = \text{cbind}(1, \dots)$$

$$X_inv = \text{solve}(t(X) \%*\% X)$$

$$Bhat = X_inv \%*\% Y \%*\% t(X) \%*\% Y$$

$$Yhat = X \%*\% Y \%*\% Bhat$$

RSS

$$n = \text{nrow}(Yhat)$$

$$ybar = \text{mean}(Y)$$

$$SSE = t(Y) \%*\% Y - t(Yhat) \%*\% Yhat$$

$$SST = t(Y) \%*\% Y - n * ybar^2$$

$$RSS = 1 - SSE/SST$$

Confidence interval

$$n = \text{nrow}(Yhat)$$

$$k = \text{ncol}(X)$$

$$S = 1/(n-k) * t(L * Y - Yhat) \%*\% (Y - Yhat)$$

$$SE = \text{sqrt}(S * X_inv[2,2])$$

$$t_crit = q(0.975, df = n-k)$$

F-test

$$\text{coeffs} = \text{as.matrix}(Bhat[2:3])$$

$$k = \text{ncol}(X)$$

$$n = \text{nrow}(Yhat)$$

$$q = \text{nrow}(\text{coeffs})$$

$$S = 1/(n-k) * t(L * Y - Yhat) \%*\% (Y - Yhat)$$

$$CI_inv = \text{solve}(X_inv[2:3,2:3])$$

$$Fstat = (t(\text{coeffs}) \%*\% CI_inv \%*\% \text{coeffs}) / (q * S)$$

$$pval = \text{pt}(Fstat, q, n-k)$$

Likelihood ratio test

$$Yhat_f = X \%*\% Y \%*\% Bhat$$

$$Full = 1/n * t(Y - Yhat_f) \%*\% (Y - Yhat_f)$$

$$X_r = X[-2:3,] \quad \rightarrow \text{take away desired predictors.}$$

$$X_inv_r = \text{solve}(t(X_r) \%*\% X_r)$$

$$Bhat_r = X_inv_r \%*\% Y \%*\% t(X_r) \%*\% Y$$

$$Yhat_r = X_r \%*\% Y \%*\% Bhat_r$$

$$\text{restricted} = 1/n * t(Y - Yhat_r) \%*\% (Y - Yhat_r)$$

$$\text{Wilks} = \det(\text{full}) / \det(\text{restricted})$$

$$n = \text{nrow}(Yhat_f)$$

$$k = \text{ncol}(X) \rightarrow \# \text{ predictors in full}$$

$$r = \text{ncol}(Y)$$

$$p = k - 1$$

$$q = k - \text{ncol}(X_r) \rightarrow \# \text{ predictors in full} - \# \text{ predictors in restricted}$$

$$\text{ChiStat} = -(n - k - 0.5 * (r - p + q + 1)) * \log(\text{Wilks})$$

$$pval = \text{pchisq}(\text{ChiStat}, df = r * q)$$

PCR

$X_{\text{-scaled}} = \text{scale}(X[, -1])$ ^{exclude intercept.}
 $\text{svd-res} = \text{svd}(X_{\text{-scaled}})$
 $\text{eig-vals} = \text{svd-res}^2$
plot elbow select k PC.

$\text{loadings} = \text{svd-res} \$ v[, 1:k]$
 $\text{scores} = X_{\text{-scaled}} \%*\% \text{loadings}$
 $X = \text{cbind}(1, \text{scores})$
 $X_{\text{-inv}} = \text{solve}(t(X) \%*\% X)$

$\text{PC-Bhat} = X_{\text{-inv}} \%*\% t(X) \%*\% Y$
 $\text{PC-Yhat} = X \%*\% \text{PC-Bhat}$

PLS

library(pls)
 $\text{pls-fit} = \text{plsrf}(Y \sim X, \text{data} = \text{df}, \text{validation} = "CV", \text{scale} = \text{TRUE})$

$\text{validation-plot}(\text{pls-fit}, \text{w.t.type} = "MSEF")$

$\text{best-ncomp} = 2$

$\text{pls-coef} = \text{coef}(\text{pls-fit}, \text{ncomp} = \text{best-ncomp}, \text{intercept} = \text{TRUE})$
 $\text{yhat-pls} = \text{predict}(\text{pls-fit}, \text{ncomp} = \text{best-ncomp}, \text{newdata} = \text{dt})$

Factor Analysis

$\text{dat-scaled} = \text{scale}(\text{dat})$
 $\text{cor-mat} = \text{cor}(\text{dat-scaled})$ $\rightarrow$ What we are trying to reproduce.

PC (better if highly correlated)

$\text{eig} = \text{eigen}(\text{cor-mat})$
 $L = \text{eig} \$ \text{vectors} \%*\% \text{diag}(\text{sqrt}(\text{eig} \$ \text{values}))$ Loading in FA = vector $\times \sqrt{\lambda_i}$, in PCA it's just vector.
 $\text{prop} = \text{sum}(\text{eig} \$ \text{values}[1:m]) / \text{sum}(\text{eig} \$ \text{values})$
 $LL = L[, 1:m] \%*\% t(L[, 1:m])$
 $h = \text{diag}(LL)$
 $\text{psi} = 1 - h$
 $L_{\text{-cor-mat}} = LL + \text{diag}(\text{psi})$

Maximum Likelihood

FA = factanal(covmat = cor_mat, factors = 2, rotation = "none")

L = FA\$loadings

LL = L %*% t(LL)

psi = FA\$uniqueness

cor_ML = LL + diag(psi)

Scores

psi_inv = diag(1/psi)

A = t(LL) %*% psi_inv %*% L

B = solve(A) %*% t(LL) %*% psi_inv

Z = scale(dat)

scores = Z %*% B

CCA

CL = cor(L1, L2)

p4 = CL\$cor[4]

→ Canonical correlations

a4 = CL\$vec[1, 4]

→

Row "a" for U

} Subst from
multi collinearity

b4 = CL\$vec[2, 4]

→

Row "b" for V

U1 = as.numeric(X1) %*% a4

→

U scores for X1

} Scores

U2 = as.numeric(X2) %*% b4

→

V scores for X2

} Scores

Canonical loadings

cor(X1, U)

cor(X2, U)

Cross loadings

cor(X1, V)

cor(X2, U)

Interpret
good!

Polak

L-pc = LC1:m

L-ml = FA\$loadings

L-pc-rot = varimax(L-pc)

L-ml-rot = varimax(L-ml)

regress

lm_fit = lm(U4 ~ X1)

cor(lm_fit\$fitted.values, U4) → should reproduce p4

to find canonical correlation.

CA

dat = smoke.

$$P = \frac{\text{dat}}{\text{sum}(\text{dat})} \quad (\text{proportion mat.})$$

row margin: $\text{rm} = \text{apply}(P, 1, \text{sum})$

col margin: $\text{cm} = \text{apply}(P, 1, \text{sum})$

$$D_r = \text{diag}(\text{rm})$$

$$D_c = \text{diag}(\text{cm})$$

$$S = \text{diag}(\text{sqrt}(1/\text{rm})) \%*\% (P - \text{rm} \%*\% \text{t}(\text{cm}) \%*\% \text{diag}(\text{sqrt}(1/\text{cm})))$$

↳ we subtract rc as if P was completely independent this would result in 0.

$$\text{svdS} = \text{svd}(S)$$

$$\text{singular} = \text{svdS}\$d \quad \rightarrow \text{inertia} = \text{sum}(\text{singular}^2)$$

$$u = \text{svdS}\$u$$

$$v = \text{svdS}\$v$$

Principal

$$p\text{-row} = \text{diag}(\text{sqrt}(1/\text{rm})) \%*\% u \%*\% \text{diag}(\text{singular})$$

$$p\text{-col} = \text{diag}(\text{sqrt}(1/\text{cm})) \%*\% v \%*\% \text{diag}(\text{singular})$$

Standard

$$s\text{-row} = \text{diag}(\text{sqrt}(1/\text{rm})) \%*\% u$$

$$s\text{-col} = \text{diag}(\text{sqrt}(1/\text{cm})) \%*\% v$$

CA package

$$\text{ca-smoke} = \text{ca}(\text{smoke})$$

$$\text{summary}(\text{ca-smoke})$$

Plot

$$\text{Symmetric plot} = \text{plot}(\text{ca-smoke}, \text{map} = "symmetric", \text{col} = c("blue", "red"))$$

$$\text{row principal} = \text{plot}(\text{ca-smoke}, \text{map} = "row principal", \text{mass} = \text{True}, \text{arrows} = (\text{TRUE}, \text{TRUE}), \text{corlib} = "absolute")$$

$$\text{col principal} = \text{plot}(\text{ca-smoke}, \text{map} = "col principal", \text{mass} = \text{True}, \text{arrows} = (\text{TRUE}, \text{TRUE}), \text{corlib} = "absolute")$$

$$\text{row-green} = \text{plot}(\text{ca-smoke}, \text{map} = "row green", \text{mass} = \text{True}, \text{arrows} = \text{TRUE}, \text{corlib} = "absolute")$$